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LIFT AND LOAD SILVERWARE BASKET

Abstract

In one embodiment, a dishwasher silverware container includes a basket with a front wall and an opposing rear wall. The basket also includes a first side wall and an opposing second side wall. The walls meet to collectively define an interior space of the basket. The basket includes at least one cam track defined within an interior surface of the basket. The dishwasher silverware container also includes a loading tray positionable about the basket. The loading tray includes a cell cover and at least one leg extending from a first side of the cell cover. Each leg has a coupling element positioned on a distal end of the leg such that the coupling element is positioned within the at least one cam track to allow vertical movement of the loading tray with respect to the basket.

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Background/Summary

TECHNICAL FIELD

[0001] The present disclosure relates to a dishwasher silverware basket, and more particularly, to a lift and load dishwasher silverware basket configured to be positioned in a loading/unloading position or a washing position.

BACKGROUND

[0002] Dishwashers are designed to clean and dry cookware, dishes, and silverware. Dishwashers typically feature an interior washing space defined by a plurality of interior walls, a door, and a pump for providing fluid communication between a water source and the interior washing space. The dishwasher may also include a plurality of racks disposed within the interior washing space configured to accept and position cookware for washing. Dishware, bakeware, glassware and the like may be supported on the rack(s) during washing, while smaller items such as silverware and utensils are placed in a basket to be washed.

[0003] In most cases, the basket is supported on a lower rack of the dishwasher. Silverware and other utensils placed in the basket are exposed to washing fluid during the washing operation. Some baskets are permanently mounted to racks, while others are removable so as to allow positioning in various locations in the dishwasher, as well as to facilitate the unloading of silverware. In this manner, the basket can be placed in a location to allow for specific placement of dishware, or removed all together to accommodate the positioning of larger items within the dishwasher.

SUMMARY

[0004] In one embodiment, a dishwasher silverware container includes a basket with a front wall and an opposing rear wall. The basket also includes a first side wall and an opposing second side wall. The walls of the basket meet to collectively define an interior space of the basket. The basket includes at least one cam track defined within an interior surface of the basket. The dishwasher silverware container also includes a loading tray positionable about the basket. The loading tray includes a cell cover and at least one leg extending from a first side of the cell cover. The at least one leg has a coupling element positioned on a distal end of the leg such that the coupling element is positioned within the at least one cam track to allow vertical movement of the loading tray with respect to the basket.

[0005] In another embodiment, a silverware container includes a basket with a front wall and an opposing rear wall. The basket also includes a first side wall and an opposing second side wall. The walls of the basket meet to collectively define an interior space of the basket. The basket further includes a first cam track defined within an interior surface of the first side wall and a second cam track defined within an interior surface of the second side wall. The silverware container also includes a loading tray positionable about the basket. The loading tray includes a cell cover; a first leg configured to be coupled with the first cam track via a first coupling element, and a second leg configured to be coupled with the second cam track via a second coupling element to enable vertical movement of the loading tray with respect to the basket. The loading tray also includes at least one hook extending approximately perpendicularly from the first leg or the second leg. The loading tray can be movable between two positions. The silverware container has a first mode where the hook interfaces with the basket to maintain the loading tray in a loading/unloading position such that the cell cover is vertically spaced above the basket and the silverware container has a second mode where the hook is disengaged from the basket and disposed within the interior space to maintain the loading tray in a washing position such that the cell cover is positioned proximate the basket to partially enclose the interior space.

[0006] Additional features and advantages of the technology described in this disclosure will be set forth in the detailed description that follows, and in part will be readily apparent to those skilled in the art from the description or recognized by practicing the technology as described in this

disclosure, including the detailed description that follows, the claims, as well as the appended drawings.

Description

BRIEF DESCRIPTION OF THE DRAWINGS

[0007] The embodiments set forth in the drawings are illustrative and exemplary in nature and not intended to limit the disclosure. The following detailed description of the illustrative embodiments can be understood when read in conjunction with the following drawings:

[0008] FIG. 1 depicts a perspective view of a loading tray and a basket, wherein the loading tray is positioned in a loading/unloading position, according to embodiments described herein;

[0009] FIG. 2 depicts a cutaway view of a basket, according to embodiments described [0010] herein; and

[0011] FIG. 3. depicts a perspective view of a loading tray comprising a first leg and a second leg, according to embodiments described herein.

[0012] Reference will now be made in greater detail to various embodiments of the present disclosure, some embodiments of which are illustrated in the accompanying drawings. Whenever possible, the same reference numerals will be used throughout the drawings to refer to the same or similar parts.

DETAILED DESCRIPTION

[0013] One of the challenges presented by current dishwasher technologies is the issue of efficiently arranging and spacing silverware within a dishwasher to achieve adequate cleaning. Conventional silverware baskets merely provide open compartments in which silverware can be placed during a dishwasher washing cycle. However, these conventional silverware baskets lead to inefficient spacing between articles of silverware that may block water flow through the basket and diminish the cleaning efficiency of the dishwasher with regard to the silverware. To overcome this spacing issue, many modern silverware baskets include a cell cover that allows a user to specifically position silverware articles within the cell cover to maintain set distances between each article. In some exemplary baskets, the proximal end of each article of silverware may be placed through the cell cover, leaving the distal end or head of the silverware exposed above the cell cover. However, such arrangements create unintended consequences, requiring a user to touch the head of each article of silverware when loading or unloading the dishwasher. As such, users are forced to come into contact with the dirty end of a used silverware head, or users are forced to handle sanitized silverware heads with potentially non-sanitized hands.

[0014] The present disclosure provides for an alternative silverware basket that includes a basket, a loading tray, a cell cover, a cam track system, and at least one hook. The silverware basket of the present disclosure creates a system in which a loading tray may be positionable about the basket and moveable relative to the basket along a cam track. The at least one hook may be positioned on the loading tray and may selectively couple with the basket to enable the loading tray to be positioned in a loading/unloading position. Furthermore, the cell cover may be positioned on an upper portion of the loading tray such that the cell cover is positioned proximate the basket when the loading tray is in the washing position and spaced apart from the basket when the loading tray is in the loading/unloading position. The cell cover may include a plurality of slots that are arranged in staggered, step-like channels that permit articles of silverware to be loaded within the slots using the handles and positioned within the cell cover so that the articles are effectively spaced to ensure optimal cleaning efficiency. Furthermore, the at least one hook may decouple from the basket and the loading tray may be positioned at least partially within the basket to place the silverware basket within a washing position.

[0015] As used hereinafter, the term “distal end” in relation to an article of silverware is used to

refer to the end of the article of silverware that is meant to contact food products and is positioned away from a user. In contrast, the term “proximal end” in relation to an article of silverware is used to refer to the end of the article of silverware that comprises a handle and is positioned near, and engaged by, the user.

[0016] Additionally, as used hereinafter, the term “distal” in relation to the basket or loading tray is used to refer to portions of the basket or loading tray that are positioned near the bottom surface of the silverware basket and/or are designed to face away from a user when seated within a dishwasher device. In contrast, the term “proximal” in relation to the basket or loading tray is used to refer to portions of the basket or loading tray that are positioned near the top of the silverware basket and/or are designed to face toward a user when seated within a dishwasher device.

Basket

[0017] Referring now to FIGS. 1 and 3, the system includes a basket **100** that is made up of a front wall **102**, a rear wall **104**, a first side wall **106**, and a second side wall **108**. The basket may optionally include a bottom surface **110**. Each of the front wall **102**, the rear wall **104**, the first side wall **106**, and the second side wall **108** extends vertically and meets two adjacent walls to define an outer perimeter of the basket **100**. In some embodiments, the front wall **102** may be positioned parallel to the rear wall **104**, but positioned as opposing sides of the basket **100**. Likewise, the first sidewall **106** may be positioned parallel to the second side wall **108**, but positioned as opposing sides of the basket **100**. In some embodiments, the front wall **102**, the rear wall **104**, the first side wall **106**, the second side wall **108**, and the bottom surface **110** may be joined with each respective adjacent wall **102**, **104**, **106**, **108** or surface **110** at a ninety degree angle. The basket **100** may define a semi-enclosed interior space **140** that is bounded on five sides by an interior surface of each of the front wall **102**, the rear wall **104**, the first side wall **106**, the second side wall **108**, and the bottom surface **110**, respectively. The interior space **140** may have an opening **142** at an upper end of the basket **100**, opposite the bottom surface **110**. In other embodiments, the interior space **140** may have an opening **142** at both an upper end of the basket **100** and a bottom end of the basket **100**.

[0018] In some embodiments the constituent walls **102**, **104**, **106**, **108** and surfaces **110** may form a basket **100** with an overall rectangular prismatic structure. The length of the opposing front wall **102** and rear wall **104** may be greater than the length of the opposing first side wall **106** and second side wall **108**, such that the opening **142** has a rectangular cross-sectional area. The dimensions of the opening **142** may be configured to substantially correspond to the dimensions of the loading tray **200** (FIG. 2) and the cell cover **202** (FIG. 2) to enable operative engagement between the basket **100** and the loading tray **200** (FIG. 2), discussed further herein. In some embodiments, each of the walls **102**, **104**, **106**, **108** may have a sufficient thickness to ensure the structural integrity of the basket **100** and create a respective upper edge **112**, **114**, **116**, **118** on an upper surface of each of the walls **102**, **104**, **106**, **108**. Each of the upper edges **112**, **114**, **116**, **118** may be flat edge, a rounded edge, or may possess additional features like a protruding lip to allow a hook **206a**, **206b** (FIG. 2) to engage and couple with the upper edge **112**, **114**, **116**, **118**. In one non-limiting embodiment, the rear wall **104**, the first side wall **106**, and the second side wall **108** may have each have a uniform upper edge **114**, **116**, **118**, respectively. However, the front wall **102** may have an upper edge **112** that includes an additional feature like a protruding lip to engage and couple with a hook **206a**, **206b** (FIG. 2) of the loading tray **200** (FIG. 2) or a retaining element **210a**, **210b** (FIG. 2) of the hook **206a**, **206b** (FIG. 2).

[0019] Referring again to FIG. 3, the basket **100** and its constituent walls **102**, **104**, **106**, **108** and surfaces **110** may be formed from a metal, a molded plastic, or any other suitable material to handle the moisture and temperature conditions of a dishwasher device. In some embodiments, the basket **100** may be formed such that each constituent wall **102**, **104**, **106**, **108** or surface **110** includes a plurality of apertures **130** therethrough to permit fluid communication through the basket **100** into the interior space **140**. In alternative embodiments, the plurality of apertures **130** may be punched

or otherwise mechanically created in one or more solid constituent walls **102**, **104**, **106**, **108** or surfaces **110**. Each of the plurality of apertures **130** may have a circular, ovular, rectangular, or otherwise suitable cross-sectional area to permit fluid communication through the aperture **130** and into the interior space **140** of the basket **100**. The plurality of apertures **130** may be uniformly spaced within each of the constituent walls **102**, **104**, **106**, **108** or surfaces **110** or the plurality of apertures **130** may be configured to be concentrated in specific portions of the basket **100** to enable and direct optimal fluid flow through the basket **100**. In some embodiments, one or more of the constituent walls **102**, **104**, **106**, **108** or surfaces **110** may possess no apertures **130**. In further embodiments, the basket **100** may further include one or more dividers extending from an interior surface of any of the walls **102**, **104**, **106**, **108** or surfaces **110**. The dividers may be positioned within the basket **100** so as to create separate compartments for the storage of silverware, while permitting the movement of the loading tray **200** (FIG. 2) within the interior space **140** of the basket **100**.

[0020] Furthermore, the basket **100** includes at least one cam track **120** defined within an interior surface of one or more of the walls **102**, **104**, **106**, **108**. The cam track **120** may have an open channel configured to accept and guide a coupling element **208a**, **208b** through the open channel of the cam track **120**. In some embodiments, the cam track **120** may be defined within an interior surface of one or more of walls **102**, **104**, **106**, **108** such that the channel of the cam track **120** does not extend beyond the interior surface of the wall **102**, **104**, **106**, **108**. In other embodiments, the cam track **120** may protrude inward, toward the interior space **140**, such that the cam track **120** extends beyond the interior surface of the wall **102**, **104**, **106**, **108**. In some embodiments, the basket **100** may include two cam tracks **120**, wherein the cam tracks **120** are positioned opposite one another in the interior surfaces of opposing walls **102**, **104**, **106**, or **108**. For instance, the two cam tracks **120** may include a first cam track **120** positioned within an interior surface of the first side wall **106** and a second cam track **120** positioned within an interior surface of the second side wall **108**. Alternatively, the two cam tracks **120** may include a first cam track **120** positioned within an interior surface of the front wall **102** and a second cam track **120** positioned within an interior surface of the rear wall **104**.

[0021] As shown in FIG. 3, each cam track **120** may include a terminal point **126** positioned proximate the bottom end of the basket **100** and an entrance point **124** positioned proximate the opening **142**. In embodiments, the terminal point **126** may be positioned proximate the bottom surface **110**. The terminal point **126** may form a closed end of the cam track **120**, whereas the entrance point **124** may be an open portion of the cam track **120** to permit entry of a coupling element **208a**, **208b** (FIG. 2) into the cam track **120**. In some embodiments, the entrance point **124** may be positioned at an upper end of the basket **100** such that the entrance point **124** to the cam track **120** is flush with the upper edge **116** of the first side wall **106**. The cam track **120** may also be angled such that the cam track **120** slopes toward the rear wall **104** as the cam track **120** progresses from the entrance point **124** toward the terminal point **126**. The terminal point **126** may be positioned proximate the rear wall **104** and the entrance point **124** may be positioned in a center of the of the first side wall **106**, wherein the distance between the entrance point **124** and the front wall **102** is smaller than the distance between the terminal point **126** and the front wall **102**. In some embodiments, the entrance point **124** may be positioned at an upper end of the basket **100** such that the entrance point **124** to the cam track **120** is flush with the upper edge **118** of the second side wall **108**. The cam track **120** may also be angled such that the cam track **120** slopes toward the rear wall **104** as the cam track **120** progresses from the entrance point **124** toward the terminal point **126**. In some embodiments, the terminal point **126** may be positioned proximate the rear wall **104** and the entrance point **124** may be positioned in a center of the of the second side wall **108**, wherein the distance between the entrance point **124** and the front wall **102** is smaller than the distance between the terminal point **126** and the front wall **102**. Furthermore, in some embodiments, the cam track **120** may also include a lateral portion **122** wherein the lateral portion

122 is a discontinuity in the otherwise linear cam track **120**. In some embodiments, the lateral portion **122** may be a horizontal segment or a substantially horizontal segment of the cam track **120** that is approximately parallel to the bottom surface **110** of the basket **100**. The lateral portion **122** of the cam track **120** may permit the coupling elements **208a**, **208b** (FIG. 2) of the loading tray **200** (FIG. 2) to move parallel relative to the bottom surface **110** to position the loading tray **200** (FIG. 2) nearer to the front wall **102**.

Loading Tray

[0022] Referring again to FIG. 2, the dishwasher silverware container further includes a loading tray **200** that is positionable about the basket **100** (FIG. 3). The loading tray **200** includes a cell cover **202**, at least one leg **204a**, **204b**, and at least one hook **206a**, **206b**.

[0023] Referring collectively to FIG. 2 and FIG. 3, the cell cover **202** may have a length and width that substantially correspond to the cross-sectional area of the opening **142** (FIG. 3) defined by the front wall **102**, the rear wall **104**, the first side wall **106**, and the second side wall **108**. In some embodiments, the cell cover **202** is positionable about the basket **100** at an upper edge **112**, **114**, **116**, **118** of the walls **102**, **104**, **106**, **108**, respectively, such that a portion of the first side (e.g., the bottom side) of the cell cover **202** may at least partially enclose the interior space **140** of the basket **100**. In other embodiments, the cell cover **202** may have a length and width that are smaller than the length and width of the basket **100** so that the cell cover **202** may be positionable within the opening **142** such that the second side (e.g., a top side) of the cell cover **202** is flush with the upper edges **112**, **114**, **116**, **118**. Alternatively, the cell cover **202** may be positionable about the basket **100** such that the cell cover **202** is positioned within the interior space **140** and below the upper edges **112**, **114**, **116**, **118**.

[0024] Referring once again to FIGS. 2 and 3, the cell cover **202** may include a front edge **212** opposite a rear edge **214** and a first side edge **216** opposite a second side edge **218**. The cell cover **202** includes a plurality of slots **240** that form open channels extending through the cell cover **202** from the first side of the cell cover **202** to the second side of the cell cover **202**. The plurality of slots **240** may be sized to accommodate a proximal end of an article of silverware while retaining a distal end of the silverware to keep the silverware's distal end in contact with the cell cover **202** during washing. In some embodiments, each of the plurality of slots **240** may have an open end proximate the first edge **212** of the cell cover **202** and a closed end positioned proximate the second edge **214**. The open end of each of the plurality of slots **240** may permit a portion of the proximal end of an article of silverware to pass into the slot **240** and position the article within the cell cover **202** for washing. In some embodiments, the plurality of slots **240** may be stepped, wherein each of the plurality of slots comprises a series of alternating longitudinal and transverse portions. Each longitudinal portion of a slot **240** extends substantially parallel to the first side edge **216** and the second side edge **218** of the cell cover **202**, whereas each transverse portion of a slot **240** extends substantially parallel to the front edge **212** and the rear edge **214** of the cell cover **202**. In some embodiments, the open end of each slot **240** may have a lateral distance to the first side edge **216** that is less than the lateral distance from the closed end of the same slot **240** to the first side edge **216**, such that the plurality of slots **240** have a staggered, diagonal arrangement within the cell cover **202**. This arrangement may maximize the length of each slot **240** and maintain adequate separation between articles of silverware positioned within the cell cover **202** during washing.

[0025] Furthermore, each leg **204a**, **204b** of the loading tray **200** may extend from a first side (e.g. a bottom side) of the cell cover **202**. Each leg **204a**, **204b** includes a coupling element **208a**, **208b** positioned on a distal end of the leg **204a**, **204b**, such that the coupling element **208a**, **208b** is positioned on the portion of the leg **204a**, **204b** that is spaced farthest from the cell cover **202**. The coupling element **208a**, **208b** is configured to be positioned within and move throughout the cam track **120** to guide the vertical movement of the loading tray **200** with respect to the basket **100**. In some embodiments, the loading tray **200** includes a first leg **204a** configured to be coupled with the first cam track **120** positioned on an interior surface of the first side wall **106** via a first coupling

element **208a**, and a second leg **204b** configured to be coupled with the second cam track **120** positioned on an interior surface of the second side wall **108** via a second coupling element **208b** to enable vertical movement of the loading tray **200** with respect to the basket **100**. The first leg **204a** may extend from a first side (e.g., a bottom side) of the cell cover **202**, specifically the first leg **204a** may extend from a portion of the cell cover **202** proximate the first side edge **216**. The second leg **204b** may extend from a first side (e.g., a bottom side) of the cell cover **202**, specifically the second leg **204b** may extend from a portion of the cell cover **202** proximate the second side edge **218**. In an embodiment, the first leg **204a** may extend from a first corner proximate the junction of the first side edge **216** and the rear edge **214** and the second leg **204b** may extend from a second corner proximate the junction of the second side edge **218** and the rear edge **214**. In non-limiting embodiments, the legs **204a**, **204b** may be angled such that the distal portion of the legs **204a**, **204b** extends beyond the footprint of the cell cover **202**.

[0026] The coupling elements **208a**, **208b** may be protrusions extending laterally from an exterior portion of the first leg **204a** and the second leg **204b** respectively, such that a first coupling element **208a** extends from the first leg **204a** toward the interior surface of the first side wall **106** and a second coupling element **208b** extends from the second leg **204b** toward the interior surface of the second side wall **108**. In some embodiments, each coupling element **208a**, **208b** may be a rolling element that is sized to fit within the open channel of the cam track **120** and move freely along the cam track **120** to allow a user to easily position the loading tray **200** with respect to the basket **100** by applying a force to the loading tray **200** along the vertical axis. The coupling element **208a**, **208b** may be selected from wheels, rollers, ball bearing systems, pins, or other suitable means to enable unimpeded, vertical movement of the loading tray **200** along the cam track **120**.

[0027] In some embodiments, as depicted in FIG. 2, the loading tray **200** may include a handle **230** to facilitate user interaction with the loading tray **200**. The handle **230** may extend vertically from a second side of the cell cover **202** opposite the first side. The handle **230** may be a U-shaped handle **230** extending from a single edge **212**, **214**, **216**, **218** of the cell cover **202**, so as to avoid blocking the silverware holding areas of the cell cover **202**. In an embodiment, the handle **230** may extend vertically from the rear edge **214** of the cell cover **202**, such that a first portion of the handle **230** extends from a first corner proximate the junction of the first side edge **216** and the rear edge **214** and a second portion of the handle **230** extends from a second corner proximate the junction of the second side edge **218** and the rear edge **214**. In alternative embodiments, the loading tray **200** may include two handles **230** extending from opposite sides of the cell cover **202**. In other embodiments, the loading tray **200** may include a single handle **230** extending from a middle portion of the first side edge **216** to a middle portion of the second side edge **218**. In still other embodiments, no handle is included.

[0028] The loading tray **200** may also include a cross member **250** extending between the first leg **204a** and the second leg **204b**. The cross-member **250** may be a longitudinal support bar that couples the first leg **204a** with the second leg **204b** to allow the first leg **204a** and the second leg **204b** to remain thin and lightweight while retaining mechanical integrity. In some embodiments, the cross member **250** extends between the first leg **204a** and the second leg **204b**, parallel to the cell cover **202** and is positioned at a distal portion of the legs **204a**, **204b**, proximate the coupling elements **208a**, **208b**. In other embodiments, the loading tray **200** may include other numbers of cross members, including, for example, zero cross members or two or more cross members **250**. In embodiments with two cross members **250**, the cross members **250** can be oriented in several configurations, including but not limited to, an orientation wherein the cross members **250** intersect one another to form X-shaped cross members extending between the first leg **204a** and the second leg **204b**. Furthermore, each leg **204a**, **204b** includes a foot **220a**, **220b**, wherein the foot **220a**, **220b** comprises the portion of the leg **204a**, **204b** that extends below the cross member **250**. In some embodiments, each coupling element **208a**, **208b** extends from each foot **220a**, **220b** of each leg **204a**, **204b**.

[0029] Still referring to FIG. 2, the loading tray **200** includes at least one hook **206a**, **206b** coupled to at least one leg **204a**, **204b**. In some embodiments, the first leg **204a** includes a first hook **206a** positioned at a distal end of the first leg **204a** and the second leg **204b** includes a second hook **206b** positioned at a distal end of the second leg **206b**, each hook **206a**, **206b** extending from each leg **204a**, **204b** parallel to the cell cover **202**. The first hook **206a** extends from the first leg **204a** toward the front wall **102** (FIG. 3), such that the first hook **206a** is vertically spaced approximately below the first side edge **216** of the cell cover **202**. The second hook **206b** extends from the second leg **204b** toward the front wall **102** (FIG. 3), such that the second hook **206b** is vertically spaced approximately below the second side edge **218** of the cell cover **202**. In some embodiments, each hook **206a**, **206b** may include a retaining element **210a**, **210b** positioned at a terminal portion of the hook **206a**, **206b** (i.e., the portion of the hook **206a**, **206b** that is positioned nearest to the front wall **102** of the basket **100**). The retaining element **210a**, **210b** may be a protrusion that extends vertically below a bottom surface of each hook **206a**, **206b** to couple each hook **206a**, **206b** with an upper edge **112**, **114**, **116**, **118** (FIG. 3) of a basket **100**. In alternative embodiments, each hook **206a**, **206b** may be a telescoping hook **206a**, **206b**, wherein each hook **206a**, **206b** includes an inner tube and an outer tube. The inner tube may be moveable along a lateral axis such that the inner tube can extend relative to the outer tube. For example, the inner tube of the telescoping hook **206a**, **206b** may extend laterally, away from the legs **204a**, **204b** and toward the front wall **102** (FIG. 3). In alternative embodiments, the outer tube may be moveable relative to the inner tube to achieve the same functionality.

Interaction Between the Basket and the Loading Tray

[0030] Referring now collectively to FIGS. 1-3, the loading tray **200** is configured to interact with the basket **100**, such that the loading tray **200** is vertically positionable relative to the basket **100**. As discussed herein, each coupling element **208a**, **208b** may engage with a cam track **120** to enable the coupling element **208a**, **208b** to move vertically within the cam track **120**. By extension, the movement of the coupling elements **208a**, **208b** within the cam tracks **120** permits the loading tray **200** to move vertically relative to the basket **100**. In some embodiments, the loading tray **200** includes a first leg **204a** that is coupled with a first cam track **120** positioned within an interior surface of the first side wall **106** via a first coupling element **208a**, and a second leg **204b** that is coupled with a second cam track **120** positioned within an interior surface of the second side wall **108** via a second coupling element **208b**, wherein the loading tray **200** can vertically extend above the basket **100** due to the movement of the coupling elements **208a**, **208b** upwardly within the cam tracks **120**. The coupling elements **208a**, **208b** may be coupled within the cam tracks **120** such that they only restrict lateral or transverse movement, but permit vertical and rotational movement within the cam tracks **120**. In some embodiments, the coupling elements **208a**, **208b** may function as pivots, wherein the loading tray **200** may freely pivot about the coupling elements **208a**, **208b** such that the loading tray **200** may tilt forward to enable each hook **206a**, **206b** extending from each leg **204a**, **204b** may interface with an upper edge **112**, **114**, **116**, **118** of the basket **100**. In some embodiments, each hook **206a**, **206b** may interface with an upper edge **112** of the front wall **102**.

[0031] The dishwasher silverware container of the present disclosure enables the loading tray **200** to be positionable relative to the basket **100** in at least two configurations or modes: a washing position and a loading/unloading position. The loading tray **200** may be positioned in the washing position when each leg **204a**, **204b** and each hook **206a**, **206b** of the loading tray **200** is fully disposed within the interior space **140** of the basket **100**, wherein the cell cover **202** is positioned about the basket **100** such that the cell cover **202** at least partially encloses the interior space **140**, as discussed herein. In the wash position, each of the coupling elements **208a**, **208b** is engaged with the cam track **120** and positioned proximate the terminal point **126** of the cam track **120** so that the loading tray **200** is positioned at its lowest vertical position.

[0032] Alternatively, the loading tray **200** may be positioned in a loading/unloading position, such

that the cell cover **202** is vertically spaced from the basket **100**, when each coupling element **208a**, **208b** is advanced vertically along each cam track **120** until each coupling element **208a**, **208b** meets the lateral portion **122** of the cam track **120**. The coupling elements **208a**, **208b** may move laterally toward the front wall **102** within the lateral portion **122** of the cam track **120**. Additionally, the loading tray **200** may pivot about each coupling element **208a**, **208b** when positioned within the lateral portion **122** such that the loading tray **200** can tilt toward the front wall **102** while the coupling element **208a**, **208b** is positioned within the cam track **120**. Due to the lateral movement and pivot of the loading tray **200**, each hook **206a**, **206b** extending approximately perpendicularly from each leg **204a**, **204b** can be positioned forward to interface with the basket **100**. In some embodiments, the loading tray **200** may pivot about the coupling elements **208a**, **208b** to lean forward relative to the basket **100** such that the bottom surface of each hook **206a**, **206b** may engage with an upper edge **112**, **114**, **116**, **118** of a wall **102**, **104**, **106**, **108** of the basket **100** to support the loading tray **200** and maintain the loading tray **200** in the loading/unloading position. In some embodiments, each hook **206a**, **206b** may interface the basket **100** due to both the lateral movement of the coupling elements **208a**, **208b** and the pivot of the loading tray **200**, while in other embodiments each hook **206a**, **206b** may engage the basket **100** due to either the lateral movement of the coupling elements **208a**, **208b** or the pivot of the loading tray **200**. In some embodiments, each hook **206a**, **206b** may further include a retaining element **210a**, **210b** to engage with and partially surround an upper edge **112**, **114**, **116**, **118** of a wall **102**, **104**, **106**, **108** of the basket **100** maintain the loading tray **200** in the loading/unloading position. In a non-limiting embodiment, each hook **206a**, **206b** may engage with the front wall **102** to maintain the basket **100** in the loading/unloading position.

[0033] Once the loading tray **200** has been loaded or unloaded, the loading tray **200** may be returned to the washing position. To re-enter the washing position, the process is reversible, wherein each hook **206a**, **206b** may be disengaged from the basket **100**, the loading tray **200** may be rotated about the coupling elements **208a**, **208b** so that the loading tray **200** is no longer leaning forward relative to the basket **100**, and the coupling elements **208a**, **208b** may be retracted back along the lateral portion **122** and down the cam track **120** until each coupling element **208a**, **208b** is positioned proximate the terminal point **126**. In so doing, each leg **204a**, **204b** and each hook **206a**, **206b** may be fully disposed within the interior space **140** of the basket **100** and the cell cover **202** is positioned about the basket **100** such that the cell cover **202** at least partially encloses the interior space **140**, as discussed herein.

[0034] It may be noted that one or more of the following claims utilize the terms “where,” “wherein,” or “in which” as transitional phrases. For the purposes of defining the present technology, it may be noted that these terms are introduced in the claims as an open-ended transitional phrase that are used to introduce a recitation of a series of characteristics of the structure and should be interpreted in like manner as the more commonly used open-ended preamble term “comprising.”

[0035] While several embodiments have been described and illustrated herein, those of ordinary skill in the art will readily envision a variety of other means and/or structures for performing the function and/or obtaining the results and/or one or more of the advantages described herein, and each of such variations and/or modifications is deemed to be within the scope of the embodiments described herein. More generally, those skilled in the art will readily appreciate that all parameters, dimensions, materials, and configurations described herein are meant to be exemplary and that the actual parameters, dimensions, materials, and/or configurations will depend upon the specific application or applications for which the teachings is/are used. Those skilled in the art will recognize, or be able to ascertain using no more than routine experimentation, many equivalents to the specific embodiments described herein. It is, therefore, to be understood that the foregoing embodiments are presented by way of example only and that, within the scope of the appended claims and equivalents thereto, embodiments may be practiced otherwise than as specifically

described and claimed. Embodiments of the present disclosure are directed to each individual feature, system, article, material, and/or method described herein. In addition, any combination of two or more such features, systems, articles, materials, and/or methods, if such features, systems, articles, materials, and/or methods are not mutually inconsistent, is included within the scope of the present disclosure.

[0036] All definitions, as defined and used herein, should be understood to control over dictionary definitions, definitions in documents incorporated by reference, and/or ordinary meanings of the defined terms.

[0037] The indefinite articles “a” and “an,” as used herein in the specification and in the claims, unless clearly indicated to the contrary, should be understood to mean “at least one.”

[0038] The phrase “and/or,” as used herein in the specification and in the claims, should be understood to mean “either or both” of the elements so conjoined, i.e., elements that are conjunctively present in some cases and disjunctively present in other cases. Multiple elements listed with “and/or” should be construed in the same fashion, i.e., “one or more” of the elements so conjoined. Other elements may optionally be present other than the elements specifically identified by the “and/or” clause, whether related or unrelated to those elements specifically identified. Thus, as a non-limiting example, a reference to “A and/or B”, when used in conjunction with open-ended language such as “comprising” can refer, in one embodiment, to A only (optionally including elements other than B); in another embodiment, to B only (optionally including elements other than A); in yet another embodiment, to both A and B (optionally including other elements); etc.

[0039] As used herein in the specification and in the claims, “or” should be understood to have the same meaning as “and/or” as defined above. For example, when separating items in a list, “or” or “and/or” shall be interpreted as being inclusive, i.e., the inclusion of at least one, but also including more than one, of a number or list of elements, and, optionally, additional unlisted items. Only terms clearly indicated to the contrary, such as “only one of” or “exactly one of,” or, when used in the claims, “consisting of,” will refer to the inclusion of exactly one element of a number or list of elements. In general, the term “or” as used herein shall only be interpreted as indicating exclusive alternatives (i.e. “one or the other but not both”) when preceded by terms of exclusivity, such as “either,” “one of,” “only one of,” or “exactly one of.” “Consisting essentially of,” when used in the claims, shall have its ordinary meaning as used in the field of patent law.

[0040] As used herein in the specification and in the claims, the phrase “at least one,” in reference to a list of one or more elements, should be understood to mean at least one element selected from any one or more of the elements in the list of elements, but not necessarily including at least one of each and every element specifically listed within the list of elements and not excluding any combinations of elements in the list of elements. This definition also allows that elements may optionally be present other than the elements specifically identified within the list of elements to which the phrase “at least one” refers, whether related or unrelated to those elements specifically identified. Thus, as a non-limiting example, “at least one of A and B” (or, equivalently, “at least one of A or B,” or, equivalently “at least one of A and/or B”) can refer, in one embodiment, to at least one, optionally including more than one, A, with no B present (and optionally including elements other than B); in another embodiment, to at least one, optionally including more than one, B, with no A present (and optionally including elements other than A); in yet another embodiment, to at least one, optionally including more than one, A, and at least one, optionally including more than one, B (and optionally including other elements); etc.

[0041] It should also be understood that, unless clearly indicated to the contrary, in any methods claimed herein that include more than one step or act, the order of the steps or acts of the method is not necessarily limited to the order in which the steps or acts of the method are recited.

[0042] In the claims, as well as in the specification above, all transitional phrases such as “comprising,” “including,” “carrying,” “having,” “containing,” “involving,” “holding,” “composed of,” and the like are to be understood to be open-ended, i.e., to mean including but not limited to.

Only the transitional phrases “consisting of” and “consisting essentially of” shall be closed or semi-closed transitional phrases, respectively.

[0043] It is to be understood that the embodiments are not limited in its application to the details of construction and the arrangement of components set forth in the description or illustrated in the drawings. The invention is capable of other embodiments and of being practiced or of being carried out in various ways. Unless limited otherwise, the terms “connected,” “coupled,” “in communication with,” and “mounted,” and variations thereof herein are used broadly and encompass direct and indirect connections, couplings, and mountings. In addition, the terms “connected” and “coupled” and variations thereof are not restricted to physical or mechanical connections or couplings.

[0044] The foregoing description of several embodiments of the invention has been presented for purposes of illustration. It is not intended to be exhaustive or to limit the invention to the precise structure, steps, and/or forms disclosed, and obviously many modifications and variations are possible in light of the above teaching.

Claims

1. A dishwasher silverware container, comprising: a basket further comprising a front wall and an opposing rear wall, and a first side wall and an opposing second side wall, wherein each of the front wall, the rear wall, the first side wall, and the second side wall comprises an interior surface such that each interior surface collectively define an interior space; at least one cam track defined within one interior surface of the basket, the cam track comprising an entrance point configured to removably receive a loading tray within the cam track; and the loading tray positionable about the basket, the loading tray further comprising: a cell cover; and at least one leg extending from a first side of the cell cover, wherein said at least one leg has a coupling element positioned on a distal end thereof such that the coupling element is positioned within the at least one cam track to allow vertical movement of the loading tray with respect to the basket.
2. The dishwasher silverware container of claim 1, wherein the at least one cam track further comprises a first cam track positioned within an interior surface of the first side wall and a second cam track positioned within an interior surface of the second side wall.
3. The dishwasher silverware container of claim 2, wherein the at least one leg further comprises a first leg that is coupled with the first cam track via a first coupling element and a second leg that is coupled with the second cam track via a second coupling element.
4. The dishwasher silverware container of claim 3, wherein each of the first cam track and the second cam track is angled such that each cam track has a terminal point positioned proximate the rear wall and the entrance point positioned in a center of the first or the second side wall so that the distance between the entrance point and the front wall is smaller than the distance between the terminal point and the front wall.
5. The dishwasher silverware container of claim 1, wherein the coupling element further comprises a rolling element selected from wheels, rollers, ball bearing systems, or pins, to permit vertical movement of the loading tray along the cam track.
6. The dishwasher silverware container of claim 1, wherein the loading tray further comprises a handle extending from a second side of the cell cover opposite the first side.
7. The dishwasher silverware container of claim 1, wherein one or more of a bottom surface, the front wall, the rear wall, the first side wall, the second side wall, and the cell cover each define a plurality of apertures therethrough to permit fluid communication through the basket to the interior space.
8. The dishwasher silverware container of claim 1, wherein the cell cover has dimensions that correspond to an area defined by the front wall, the rear wall, the first side wall, and the second side wall, such that the cell cover is positionable at an upper edge of each of the front wall, the rear

wall, the first side wall, and the second side wall to partially enclose the interior space.

9. The dishwasher silverware container of claim 1, wherein the cell cover defines a plurality of slots extending from a first edge to a second edge opposite the first edge, wherein each slot in the plurality of slots is open proximate the first edge and closed proximate the second edge.

10. The dishwasher silverware container of claim 9, each slot comprising a series of alternating longitudinal and transverse portions, wherein each longitudinal portion extends parallel to a first side edge and a second side edge of the cell cover and each transverse portion extends perpendicular to the longitudinal portions.

11. A silverware container, comprising: a basket comprising a front wall and an opposing rear wall and a first side wall and an opposing second side wall, wherein each of the front wall, the rear wall, the first side wall, and the second side wall comprises an interior surface such that each interior surface collectively define an interior space; a first cam track defined within the interior surface of the first side wall and a second cam track defined within the interior surface of the second side wall, each of the first cam track and the second cam track comprising an entrance point configured to removably receive a loading tray within the cam track; and the loading tray positionable about the basket, the loading tray further comprising: a cell cover; a first leg configured to be coupled with the first cam track via a first coupling element, and a second leg configured to be coupled with the second cam track via a second coupling element, to enable vertical movement of the loading tray with respect to the basket; and at least one hook extending approximately perpendicularly from the first leg or the second leg, wherein the loading tray can be movable between two positions, wherein in a first mode the at least one hook interfaces with the basket to maintain the loading tray in a loading/unloading position such that the cell cover is vertically spaced above the basket, and in a second mode the at least one hook is disengaged from the basket and disposed within the interior space to maintain the loading tray in a washing position such that the cell cover is positioned proximate the basket to partially enclose the interior space.

12. The silverware container of claim 11, wherein: the at least one hook comprises a first hook and a second hook, the first leg further comprises the first hook positioned at a distal end of the first leg, extending perpendicularly from the first leg toward the front wall, and the second leg comprises the second hook positioned at a distal end of the second leg, extending perpendicularly from the second leg toward the front wall.

13. The silverware container of claim 11, wherein each of the first cam track and the second cam track are angled such that each cam track has a terminal point positioned proximate the rear wall and the entrance point positioned in a center of the first side wall or the second side wall so that the distance between the entrance point and the front wall is smaller than the distance between the terminal point and the front wall.

14. The silverware container of claim 13, wherein each cam track further comprises a lateral portion that permits the loading tray to move approximately parallel relative to a bottom surface to position each hook proximate the front wall.

15. The silverware container of claim 11, wherein the loading tray is free to pivot about the first coupling element or the second coupling element such that each hook is configured to interface with an upper edge of the front wall to fix the loading tray in the loading/unloading position.

16. The silverware container of claim 11, wherein each hook is a telescoping hook comprising an inner tube and an outer tube, such that the inner tube is configured to extended relative to the outer tube to interface with the front wall.

17. The silverware container of claim 11, wherein each hook comprises a retaining element extending from a terminal end of the hook configured to interact with an upper edge of the front wall.

18. The silverware container of claim 11, the loading tray further comprising a cross member extending between the first leg and the second leg and each of the first leg and the second leg further comprises a foot below the cross member, wherein each of the first coupling element and

the second coupling element extends from each foot.

19. The silverware container of claim 11, wherein each of the first coupling element and the second coupling element comprises a rolling element selected from wheels, rollers, ball bearing systems, pins, or other suitable means to permit movement within each cam track.

20. A silverware container, comprising: a basket comprising a front wall and an opposing rear wall and a first side wall and an opposing second side wall, wherein each of the front wall, the rear wall, the first side wall, and the second side wall comprises an interior surface such that each interior surface collectively define a basket interior space; a first cam track defined within the interior surface of the first side wall and a second cam track defined within the interior surface of the second side wall, each of the first cam track and the second cam track comprising an entrance point configured to removably receive a loading tray within the cam track, wherein each of the first cam track and the second cam track are angled such that each cam track has a terminal point positioned proximate the rear wall and the entrance point positioned in a center of the first side wall or the second side wall so that the distance between the entrance point and the front wall is smaller than the distance between the terminal point and the front wall and each cam track further comprises a lateral portion that permits the loading tray to move approximately parallel relative to a bottom surface to position a first hook and a second hook proximate the front wall; and a loading tray positionable about the basket, the loading tray further comprising: a cell cover; a first leg configured to be coupled with the first cam track via a first coupling element, and a second leg configured to be coupled with the second cam track via a second coupling element, to enable vertical movement of the loading tray with respect to the basket; and the first hook extending approximately perpendicularly from the first leg and the second hook extending approximately perpendicularly the second leg, wherein the loading tray can be movable between two positions, wherein in a first mode the first hook and the second hook interface with the basket to maintain the loading tray in a loading/unloading position such that the cell cover is vertically spaced above the basket, and in a second mode the first hook and the second hook are disengaged from the basket and disposed within the basket interior space to maintain the loading tray in a washing position such that the cell cover is positioned proximate the basket to partially enclose the basket interior space.
